Sensitivity of BioEmu Stability Benchmark Estimates to Sequential Post-hoc Structural Filtering

Fixed-archive technical note | 24 October 2026 | Sho Miyazaki · Independent researcher, Tokyo, Japan

Abstract

We evaluated sensitivity to sequential post-hoc structural filters in a saved BioEmu stability benchmark with 154 systems and 112 mutant–WT pairs. The archived A ensemble (302,800 frames) was reduced to B0 (298,188) by official geometry screening and then to Bω (272,381) by a fixed peptide-ω screen. Predicted paired ΔΔG shifts of magnitude at least 0.1 kcal/mol occurred in 19/112 pairs for A→B0 and 40/112 pairs for B0→Bω, with 12 in common. Descriptive MAE relative to the same experimental references was 0.702595, 0.696142 and 0.707681 kcal/mol for A, B0 and Bω. System-level foldedness and ΔG responses illustrate amplification or cancellation in mutant-minus-WT estimates. These results document fixed-archive sensitivity, not generally improved predictions or restored equilibrium populations.

1. Background and research questions

BioEmu produces protein structural ensembles and offers an established pathway to soft foldedness and mutation-related stability estimates [1]. This technical record asks how the established geometry screen and subsequent fixed peptide-backbone ω screen alter estimates within one archived benchmark. It does not introduce a new energy estimator, packing algorithm or optimized structural-quality threshold.

2. Data and frozen methods

The fixed archive includes 42 WT and 112 mutant system entries, yielding 112 mutant–matched-WT comparisons with shared WT references. A includes 302,800 saved frames; B0 is the official geometry-pass subset; Bω is a subset of B0 after cis/trans-aware ω evaluation using CA(i)–C(i)–N(i+1)–CA(i+1) and strictly greater than 30° deviation from the appropriate planar reference. Cis is not automatically rejected. The frozen calculation uses explicit FNCSettings(contact_lambda=1.2, contact_delta=0.0), the fixed soft-foldedness sigmoid, a 295-K logit transformation to predicted ΔG, and ΔΔG = ΔG(mutant) − ΔG(matched WT). The 0.1-kcal/mol absolute shift is a descriptive reporting threshold, not statistical or experimental significance.

3. Results: retention and foldedness

The official geometry filter removed 4,612/302,800 frames (1.523%); the additional ω screen removed 25,807/298,188 frames (8.655% of B0). No unknown frames were recorded in the frozen X2 arrays. Across 154 systems, Bω minus B0 mean soft-foldedness changes had median +0.001850 and range −0.011552 to +0.045725; 131 systems increased and 23 decreased. These values condition on retained saved frames and are not corrections to an equilibrium population.

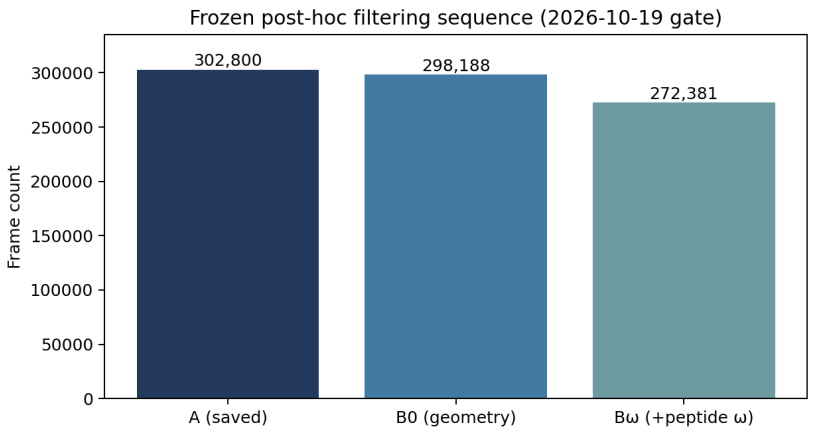


Figure 1. Successive retained frame counts in the saved A ensemble, official geometry subset B0 and additional peptide-ω subset Bω.

4. Results: system-level ΔG sensitivity

System-level changes reflect arithmetic means of retained soft-foldedness contributions followed by the fixed nonlinear logit transform; removal fractions alone do not determine the signed ΔG shift. Representative B0→Bω changes were −0.570955 kcal/mol for 1EM7_L7A__V39G, +0.130724 for 1TG0, −0.239841 for 1TG0__E1N and −0.438085 for 6FVC. For the matched 1TG0__E1N minus 1TG0 pair, opposite-signed system ΔG changes generated a paired shift of approximately −0.370565 kcal/mol. Exact per-system values are provided in SI and CSV.

5. Results: paired ΔΔG and cross-stage comparisons

Among 112 comparisons, 19 had |ΔΔG(A→B0)| ≥ 0.1 kcal/mol and 40 had |ΔΔG(B0→Bω)| ≥ 0.1 kcal/mol. The intersection was 12; their union was 47, while the cumulative A→Bω absolute shift crossed the threshold in 44. This difference is consistent with signed amplification and cancellation. Descriptive MAEs were 0.702595, 0.696142 and 0.707681 kcal/mol; between B0 and Bω, absolute errors improved for 54 paired comparisons and worsened for 58.

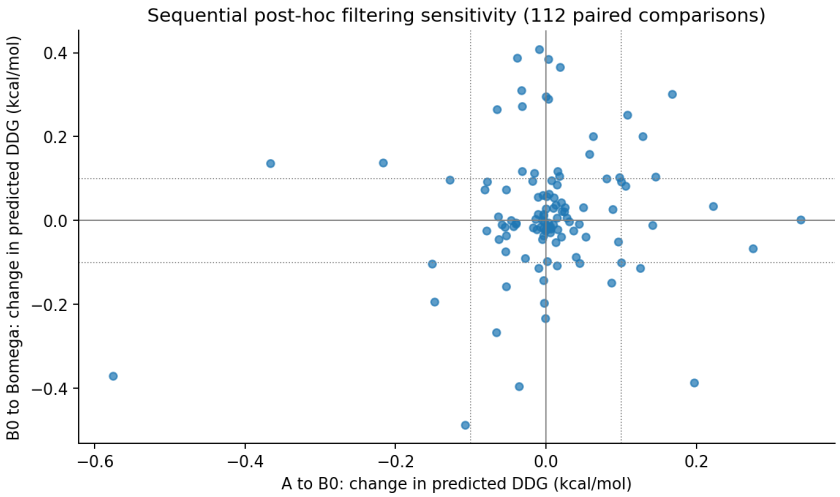


Figure 2. Paired changes in predicted ΔΔG (kcal/mol) for 112 mutant–WT comparisons: A→B0 against B0→Bω. Dotted threshold guides indicate the descriptive ±0.1-kcal/mol shifts.

6. Discussion

The retained conformation set, rather than the frame-removal percentage alone, determines how soft foldedness and system-level ΔG estimates change. Shared-WT references can cancel or amplify mutant changes in paired ΔΔG. The two structural screens affect overlapping but distinct pairs, so the union of pairs exceeding separate stage thresholds differs from the cumulative A→Bω count. Small descriptive MAE differences cannot establish a generally preferred filter. The scientific value here is a reproducible correspondence among archive, masks, system estimates and pairwise endpoints.

7. Sensitivity and limitations

The B0 ω evaluation comprised 14,922,082 evaluable bond–frame observations, of which 27,329 were classified as twisted. There were 292 frames within ±0.02° of the primary cutoff; two existing boundary alternatives gave 39/112 threshold-exceeding comparisons, compared with the frozen primary 40/112. The three under-70 systems use target-frame—not B0-frame—denominators. The saved A ensemble is not the unknown unsaved generative population. Shared-WT pairs are dependent, no random-removal control was undertaken, and filter selection is not a thermodynamic reweighting correction. All 154 saved NPZ arrays underwent arithmetic QA; independent original-coordinate QA was limited to 10 selected systems/2,700 frames. Original historical dependency versions remain incompletely documented.

8. Conclusions

In this fixed 154-system BioEmu benchmark, two sequential structural screens altered predicted stability estimates in partly overlapping subsets of mutant–WT comparisons. System-level foldedness and ΔG responses account for amplification and cancellation in paired estimates. No generalized performance improvement, restored equilibrium sampling or filter-specific causal benefit is established; the deliverable is a traceable fixed-archive sensitivity record.

9. Data availability and declarations

The companion package includes 154 saved NPZ members, reconciled system and 112-pair tables, ω results, threshold diagnostics, figures, archived scripts and SHA-256 checksums. The upstream benchmark is Zenodo record 15672282. Author: Sho Miyazaki (Independent researcher, Tokyo, Japan). Funding: no external funding declared for these independent analyses. Competing interests: none declared. Acknowledgments: authors and maintainers of BioEmu and upstream benchmarks are credited by citation; no additional personal acknowledgments are asserted. AI disclosure: ChatGPT assisted with research planning, code and data checks, manuscript and figure preparation and QC; the author directed the research and remains responsible for interpretation and release. Author-owned writing and figures: CC BY 4.0; original owned code: MIT; third-party data retain separate terms. Public repository URL and DOI are pending deposition.

References

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[3] Microsoft. BioEmu benchmarks: source code and data release. https://github.com/microsoft/bioemu-benchmarks (pinned source and Zenodo 15672282).